33. Multi-Channel Decimation for Massive MIMO

Simulate multi-channel decimation in Massive MIMO receiver and analyze computational reduction. (25) (BL4) (WK4)

Given Data:

Number of Channels = 16
Sampling Rate = 122.88 MHz
Decimation Factor = 4

Tasks:

Generate multi-channel signals.
Design a common decimation filter.
Perform decimation.
Analyze computational reduction.
Compare channel spectra.
Discuss implementation benefits.

MATLAB Code:

```
clc;
clear;
close all;

%% Multi-Channel Decimation for Massive MIMO
% Given Data
Nch = 16;                % Number of channels
Fs = 122.88e6;           % Sampling frequency = 122.88 MHz
M = 4;                   % Decimation factor
Fout = Fs/M;             % Output sampling frequency

%% 1. Generate Multi-Channel Signals

T = 1e-4;                % Simulation time
t = 0:1/Fs:T-1/Fs;
N = length(t);

% Allocate input signal matrix
x = zeros(Nch,N);
```

```
% Different frequency for each channel
channelFreq = linspace(5e6,40e6,Nch);

for ch = 1:Nch
   % Multi-channel sinusoidal signal
   x(ch,:) = sin(2*pi*channelFreq(ch)*t);

   % Add small noise
   x(ch,:) = x(ch,:) + 0.05*randn(1,N);
end

fprintf('Number of Channels       = %d\n',Nch);
fprintf('Input Sampling Rate      = %.2f MHz\n',Fs/1e6);
fprintf('Decimation Factor        = %d\n',M);
fprintf('Output Sampling Rate     = %.2f MHz\n\n',Fout/1e6);
```

%% 2. Design a Common Decimation Filter

```
% FIR low-pass filter
filterOrder = 63;

% Cutoff frequency
Fc = Fout/2;

% Normalized cutoff frequency
Wn = Fc/(Fs/2);

% Common filter used for all channels
b = fir1(filterOrder,Wn);

fprintf('Common FIR Filter Order  = %d\n',filterOrder);
```

%% **3. Perform Filtering and Decimation**

```
y = zeros(Nch,ceil(N/M));
```

```
for ch = 1:Nch

    % Filter the signal
    filteredSignal = filter(b,1,x(ch,:));

    % Downsample by factor of 4
    y(ch,:) = filteredSignal(1:M:end);

end
```

%% 4. Computational Reduction Analysis

```
% Without decimation:
% Every input sample is processed at Fs

% With decimation:
% Only 1 out of every M samples is retained

originalSamples = N * Nch;
decimatedSamples = size(y,2) * Nch;

reductionPercentage = ...
    (1 - decimatedSamples/originalSamples)*100;

fprintf('\n----- Computational Analysis -----\n');
fprintf('Original samples processed = %d\n',originalSamples);
fprintf('After decimation           = %d\n',decimatedSamples);
fprintf('Computational reduction    = %.2f %%\n',reductionPercentage);
```

%% 5. Plot Input Signals of First 4 Channels

```
figure;

for ch = 1:4
    subplot(4,1,ch);
    plot(t(1:1000)*1e6,x(ch,1:1000));
    grid on;
```

```
    xlabel('Time (\mus)');
    ylabel('Amplitude');
    title(['Input Signal - Channel ',num2str(ch)]);
end
```

%% 6. Plot Output Signals of First 4 Channels

```
t_out = (0:size(y,2)-1)/Fout;

figure;

for ch = 1:4
    subplot(4,1,ch);
    plot(t_out(1:min(1000,length(t_out)))*1e6,...
        y(ch,1:min(1000,length(t_out))));
    grid on;
    xlabel('Time (\mus)');
    ylabel('Amplitude');
    title(['Decimated Signal - Channel ',num2str(ch)]);
end
```

%% 7. Compare Channel Spectra

```
figure;

for ch = 1:Nch

    % FFT of decimated signal
    Y = fft(y(ch,:));
    L = length(Y);

    P2 = abs(Y/L);
    P1 = P2(1:floor(L/2)+1);

    f = (0:floor(L/2))*Fout/L;

    subplot(4,4,ch);
```

```
    plot(f/1e6,20*log10(P1+1e-12));
    grid on;

    xlabel('Frequency (MHz)');
    ylabel('Magnitude (dB)');
    title(['Channel ',num2str(ch)]);

end

sgtitle('Spectra of 16 Decimated MIMO Channels');
```

%% 8. Display Channel Frequencies

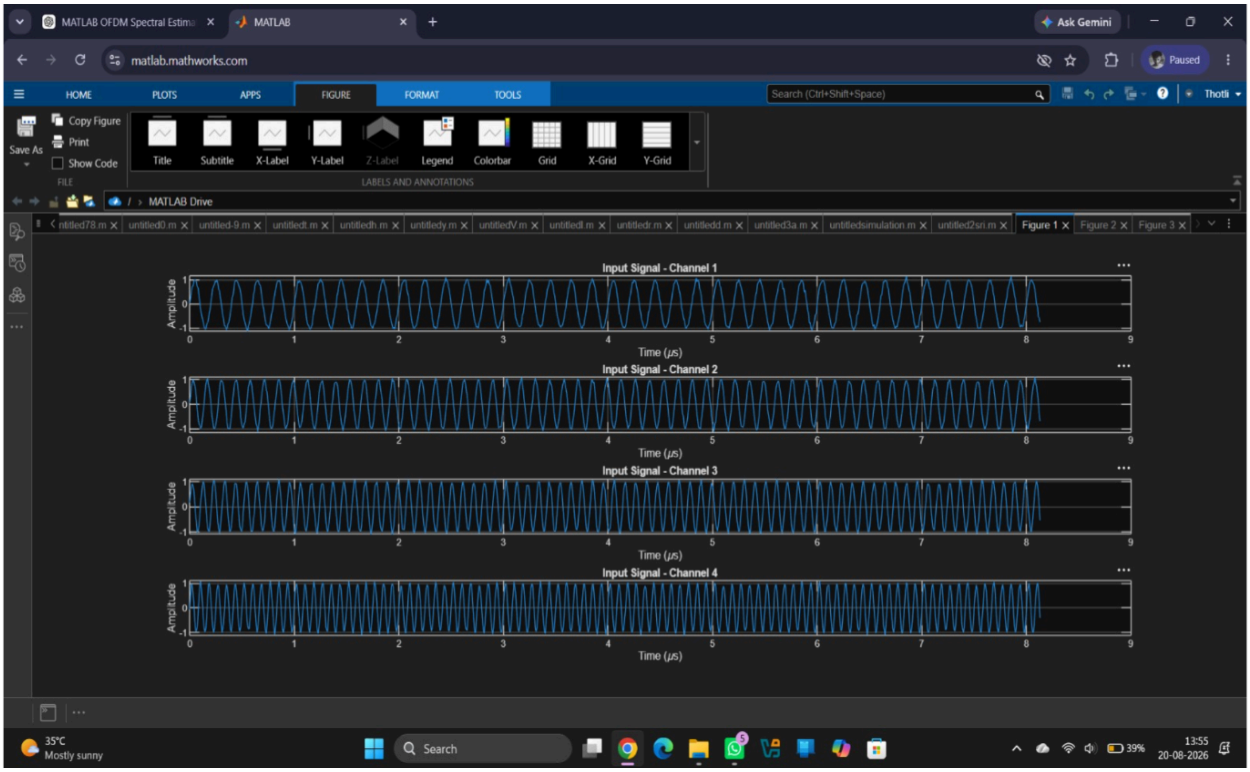
```
fprintf('\n----- Channel Frequency Information -----\n');

for ch = 1:Nch
    fprintf('Channel %2d : %.2f MHz\n', ...
        ch,channelFreq(ch)/1e6);
end
```

%% 9. Computational Benefit

```
fprintf('\n----- Implementation Benefits -----\n');
fprintf('1. Number of channels       : %d\n',Nch);
fprintf('2. Decimation factor        : %d\n',M);
fprintf('3. Sampling rate before     : %.2f MHz\n',Fs/1e6);
fprintf('4. Sampling rate after      : %.2f MHz\n',Fout/1e6);
fprintf('5. Data-rate reduction      : %.2f %%\n',reductionPercentage);
fprintf('6. Common filter is shared among all channels.\n');
fprintf('7. Lower processing and memory requirements.\n');
fprintf('8. Reduced communication bandwidth.\n');
```

OUTPUT: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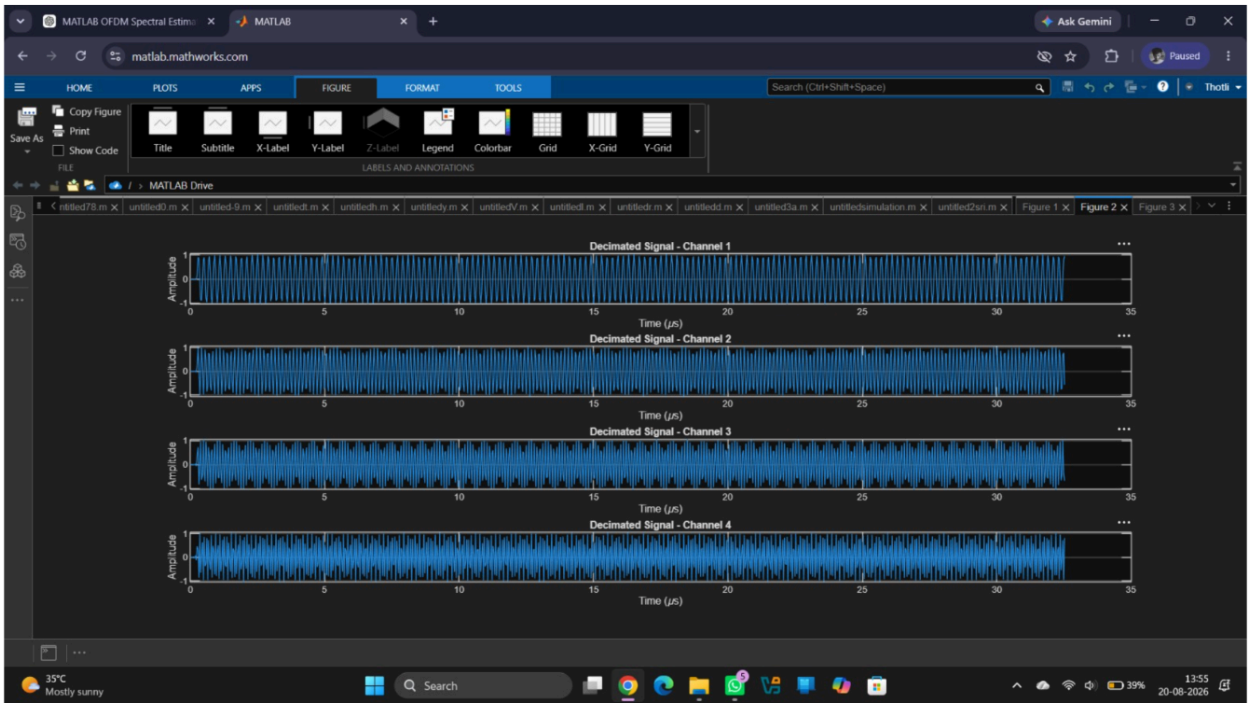
LABELS AND ANNOTATIONS
MATLAB Drive
Figure 1
Figure 2
Figure 3
Input Signal - Channel 1
Input Signal - Channel 2
Input Signal - Channel 3
Input Signal - Channel 4
Amplitude
Time (μs)
35°C
Mostly sunny
Search
39%
13:55
20-08-2026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LABELS AND ANNOTATIONS
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Figure 1
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Decimated Signal - Channel 1
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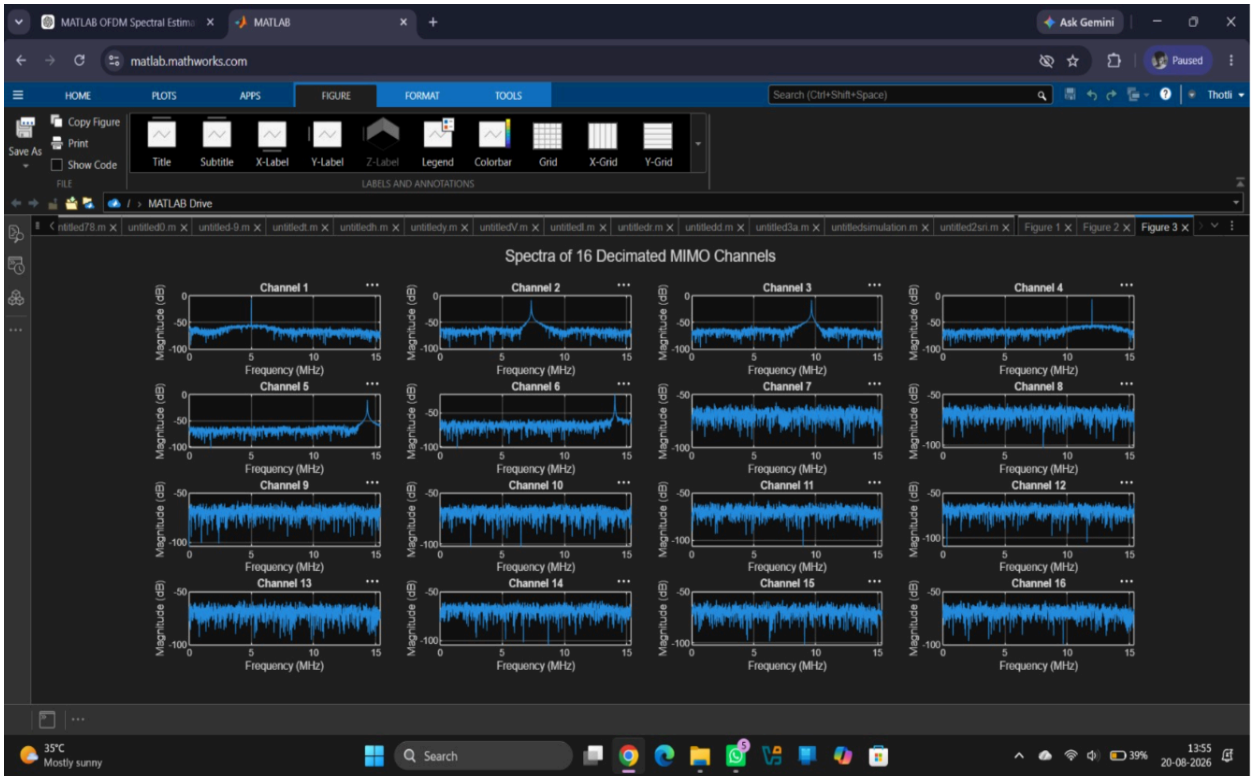
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Figure 1
Figure 2
Figure 3
Spectra of 16 Decimated MIMO Channels
Channel 1
Channel 2
Channel 3
Channel 4
Channel 5
Channel 6
Channel 7
Channel 8
Channel 9
Channel 10
Channel 11
Channel 12
Channel 13
Channel 14
Channel 15
Channel 16
Magnitude (dB)
Frequency (MHz)
35°C
Mostly sunny
Search
39%
13:55
20-08-2026